

CS5002NI Software Engineering

20% Group Coursework

AY 2025-2026

Credit: 30

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Milestone Due Date: 2026/01/19 (6:00pm)

Milestone Submission Date: 2026/01/19 (5:30PM)

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1 Introduction

Gokyo Bistro is a renowned restaurant situated in Jhamsikhel, Lalitpur, known for its friendly environment and varied menu items that attract hundreds of customers every day. Due to the rise in customer volume day by day, restaurants are now facing various challenges in managing table reservations and cancellations, along with order processing, which will further increase with the demand for home delivery services. Presently, most of the activities depend directly on manpower, which is generally time-consuming and error-prone. Thus, management is interested in developing an online system to manage daily activities effectively and provide good service to customers. The proposed system will support the following: customer registration and login, advance table booking with secure payment, reservation rules and cancellation, beverage preferences, online home delivery ordering, automated billing, report generation, menu management, customer rating and feedback, and managing promotional offers thereby increasing customer satisfaction along with operational efficiency.

2 Group Task

2.1 Business Case

A. Business Problem

Currently Gokyo Bistro experiences the following problems:

- Overcrowding, maximum occupancy issues during rush hour.
- Manual, semi-manual reservation and also order handling leading to errors and delays.
- Facing difficulty in managing home delivery requests and payment confirmations.
- Lack of automation of billing, reporting, and decision support (revenue prediction, menu recommendations) (yung, 2026).

B. Justification for Developing the System

The following points justified why the System must be developed:

- It provides centralized control over customers data, payments, orders and bookings.
- Automating these things minimizes human errors and manual tasks.
- It enables run-time updates on table availability and status of order.

- It supports various online payments get way, which are essential in modern industries for services.
- For enhancing customer experiences.
- For increasing operational efficiency.
- For expending delivery services
- For using data for business decision making.

C. Proposed Solution Overview

The proposed solution is a web-based, centralized, and automated Online Restaurant Management System that will support the core operations of Gokyo Bistro. The system will provide an integrated platform where customers and staff can interact with restaurant services digitally. It will include following things:

- User registration and login.
- Table booking with advance payment.
- Reservation cancellation with refund rules.
- Home delivery ordering for members.
- Menu browsing and updates.
- Automated billing and receipts.
- Ratings and feedback.
- Promotional offers and discounts.
- Financial reporting and basic prediction.

Primary Users of the system:

- **Visitor:** Can register, browse menu, book tables, and give feedback.
- **Members:** Can login, book tables, place delivery orders, and manage preferences.
- **Admin (Staff/Manager):** Manage menu, bookings, bills, reports, and offers.

D. Objective of the System

The following things are the main objective of the System to make operation faster, more accurate, and more customer centric:

Operational Objectives

- To automate table reservations and prevent overbooking.
- To streamline home delivery order processing.
- To reduce manual paperwork and errors.

User-Oriented Objectives

- To provide convenient online booking and payment.
- To enable feedback and ratings for service improvement.
- To offer a simple and user-friendly interface.

Management Objectives

- To generate accurate financial reports.
- To manage menu items and promotional offers.
- To support basic revenue prediction and menu recommendations.

E. Expected Benefits**Tangible Benefits**

- Lower operational costs due to automation.
- Faster order and payment processing.
- Improved utilization of tables.
- More accurate financial records.

Intangible Benefits

- Increased trust due to transparent payments and refunds.
- Improved restaurant image as a modern business.
- Higher customer satisfaction and loyalty.

Benefits to Stakeholders

- Customers: Convenience, faster service, online access.
- Staff/Admin: Reduced workload, easier management.
- Management: Better control, insights, and decision-making.

Long-Term Benefits

- It increases Business scalability.
- It increases data-driven growth and sustainability.

F. Budget of the Project

This is the approximate estimation for this project.

Table 1 Budget of the Project

Cost Components	Estimated Cost (NPR)
Development (design, coding, testing)	120,000 – 180,000
Server and Hosting (annual)	20,000 – 30,000
Database and Tools	10,000 – 15,000
Maintenance and Support (annual)	25,000 – 40,000
Training and Deployment	10,000 – 15,000

Estimated Total: NPR 185,000 – 280,000

Assumption

- This development done by a small team.
- Use of open-source tools.
- Cloud hosting with basic configuration.
- Budget excludes large-scale marketing.

G. Risks and Constraints**Technical Risks**

- Software bugs and performance issues.
- Integration issues with payment gateway.
- Data loss or security vulnerabilities.

Financial Risks

- Budget Overruns.
- Unexpected infrastructure costs.

Time Constraints

- Limited academic timeline for development.
- Delays due to complexity of features.

Human Risks

- Resistance from staff to adopt new system.
- Limited technical skills of users.

Legal and Operational Constraints

- Compliance with online payment regulations.
- Data privacy and security requirements.

Mitigation Strategies

- Incrementing testing and reviews.
- Backup and security measures.
- Realistic scheduling and scope control.
- User training sessions.

H. Timeline with Major Milestones

This project is expected to be completed within 12 – 14 weeks.

Table 2 Timeline with Major Milestones

Phase	Duration	Deliverables
Requirement Analysis	2 weeks	SRS, use cases
System Design	2 weeks	DFDs, ERD, UI mockups
Development	5-6 weeks	Functional system modules
Testing	2 weeks	Test cases, bug fixes
Deployment	1 week	Deployed system, user guide
Documentation and Review	1 week	Final report, presentation

3 SRS (Software Requirements Specification)

3.1 System Overview

The system allows Admin, Visitors, and Members to interact with restaurant services: table booking with advance payment, cancellation, beverage reservation, home delivery ordering (members), billing, reporting, menu updates, ratings, and promotional offers (Krüger, 2025).

3.2 Functional Requirements (FR)

3.2.1 User Management (Login & Registration):

FR1 The system shall allow users to register as **Visitor** or **Member**.

FR2 The system shall allow **Admin** and **Members** to login using credentials.

FR3 The system shall validate login credentials and grant access based on user role.

FR4 The system shall allow users to logout securely.

3.2.2 Table Booking:

FR5 The system shall allow Visitors/Members to check table availability for a selected date/time.

FR6 The system shall allow Visitors/Members to reserve a table for a particular date/time.

FR7 The system shall require an **advance payment of Rs. 1200** to confirm a booking.

FR8 The system shall generate a booking confirmation after successful payment.

3.2.3 Reservation Cancellation:

FR9 The system shall allow cancellation of reservation before the final day.

FR10 If reservation is cancelled on the same day, the system shall deduct **Rs. 500** from advance payment before refund.

FR11 The system shall update booking status to “Cancelled” and store refund details.

3.2.4 Set Beverage Choices:

FR12 The system shall allow Visitors to optionally select beverage preferences during or after booking.

FR13 The system shall allow Members to reserve their wine/whiskey choices from the available list.

FR14 The system shall store beverage choices associated with a booking/user.

3.2.5 Place Order for Home Delivery (Members only):

FR15 The system shall allow Members to place a home delivery order.

FR16 The system shall allow online payment for delivery orders.

FR17 The system shall generate an order confirmation and expected delivery time.

3.2.6 Generate Report (Admin):

FR18 The system shall allow Admin to generate financial reports (daily/weekly/monthly).

FR19 The system shall provide a mechanism to predict future revenue (based on historical sales).

FR20 The system shall recommend menu items based on sales trends.

3.2.7 Bill Generation (Admin):

FR21 The system shall allow Admin to generate bills for customers.

FR22 The system shall generate receipts for payments.

3.2.8 Update Menu (Admin):

FR23 The system shall allow Admin to add menu items.

FR24 The system shall allow Admin to update menu items.

FR25 The system shall allow Admin to delete menu items.

3.2.9 Rate Experience:

FR26 The system shall allow Visitors/Members to rate food, staff behavior, and ambience.

FR27 The system shall allow users to submit feedback text along with rating.

3.2.10 Post Offers & Schemes (Admin):

FR28 The system shall allow Admin to post offers and discount schemes.

FR29 The system shall show active offers to users.

3.3 Non-Functional Requirements (NFR)

3.3.1 Security Requirements:

NFR1 The system shall store all user password using **one-way cryptographic** hashing and shall never store passwords in plain text.

NFR2 The system shall enforce **role-based** authorization to restrict access to system features based on user roles.

NFR3 The system shall long all payment and refund transactions with transactions ID, timestamp, amount, and status for audit purposes.

3.3.2 Performance Requirements:

NFR4 The system shall display table availability results within **3 seconds** for up to **100** concurrent users.

NFR5 The system shall generate Standard financial reports within **10 seconds** for up to **12 months** of data.

NFR6 The system shall process payments confirmation within **5 seconds** of receiving a gateway response.

3.3.3 Reliability and Data Integrity:

NFR7 The system shall prevent double booking of the same table for overlapping time slots.

NFR8 The system shall confirm reservations or delivery orders only when payment status is marked as "Success".

NFR9 The system shall maintain consistency between billing records and payment transaction at all times.

3.3.4 Usability Requirements:

NFR10 The system shall provide an intuitive user interface enabling first-time users to complete booking or orders without training.

NFR11 The system shall display clear and informative error messages for invalid input, payments failures, or rule violations.

3.3.5 Maintainability and Scalability:

NFR12 The system shall follow a modular architecture separating booking, delivery, billing, reporting, and user management modules.

NFR13 The system shall support future enhancements without requiring major redesign components.

3.3.6 Availability Requirements:

NFR14 The system shall maintain at least **90% availability**, excluding scheduled maintenance.

NFR15 The system shall recover from failures without loss of confirmed reservations, orders, or payment data.

3.4 Design & Implementation Constraints

- The system must support three user types: **Admin, Visitors, Members**.
- Online payments must be supported for advance booking payment and delivery order payment.
- Advance payment (**RS 1200**) is mandatory for table bookings confirmation.
- Same day cancellation must deduct **RS. 500** from the advance payment.

3.5 External Interfaces Required

3.5.1 User Interface

- Web/mobile-friendly pages for: Login/Register, Booking, Delivery Order, Ratings, Offers
- Administrative dashboards for menu management, billing, reporting, and promotions.

3.5.2 Hardware Interfaces

Client-side devices

- Smartphones, tablets, and PCs/laptops with modern browsers for users and admins.

Server-side infrastructure

- Application server + database server (can be cloud VM or hosted server).
- Storage for: database, logs, backups, and (if used) report exports (PDF/CSV).

Admin hardware

- USB / Network thermal printer (typical POS printer).
- Output format: 80mm receipt layout (order/bill ID, items, totals, payment status).

3.5.3 Software Interfaces

Database / storage

- RDBMS such as MySQL or PostgreSQL.
- Key data entities stored: Users (Admin/Visitor/Member), Reservations, Payments, Orders, Menu Items, Offers, Ratings, Refund records.

Required API capabilities:

- Create payment request (amount, purpose, reference ID).
- Verify payment status callback/webhook.
- Refund initiation + refund status tracking.

Reporting/analytics interface

- Export reports to PDF/CSV for admin download.
- Revenue prediction & menu recommendation module may use local analytics.

3.5.4 Communication Interfaces

Client ↔ Server

- All communication over HTTPS with TLS encryption.
- API style: RESTful endpoints using JSON request/response.
- Session handling: secure cookies or token-based authentication (role-based access).

Server ↔ External services

- Payment gateway communication:
 - HTTPS API calls + webhook/callback endpoint for payment confirmation.
 - Timeout/retry policy (e.g., retry payment verification if gateway delays).
- Delivery/notification API calls:
 - HTTPS API calls to SMS/email provider and delivery provider.

Security & operational interface requirements

- Input validation on all endpoints (prevent SQL injection/XSS).
- Rate limiting on login/payment endpoints to reduce abuse.
- Audit logs for admin actions (menu updates, bill generation, refund processing).

4 DFD's (level 0, level 1 and level 2)

4.1 Context Diagram / Level 0 DFD (Environmental Model)

External Entities

- Visitor
- Member
- Admin
- Payment Gateway

Single Process

- P0: Gokyo Bistro Online System

Main Data Flows

- Registration/Login data
- Reservation request + confirmation
- Advance payment request/response
- Cancellation request + refund details
- Delivery order + payment + confirmation
- Menu updates
- Bill/receipt
- Reports
- Ratings/feedback
- Offers/schemes

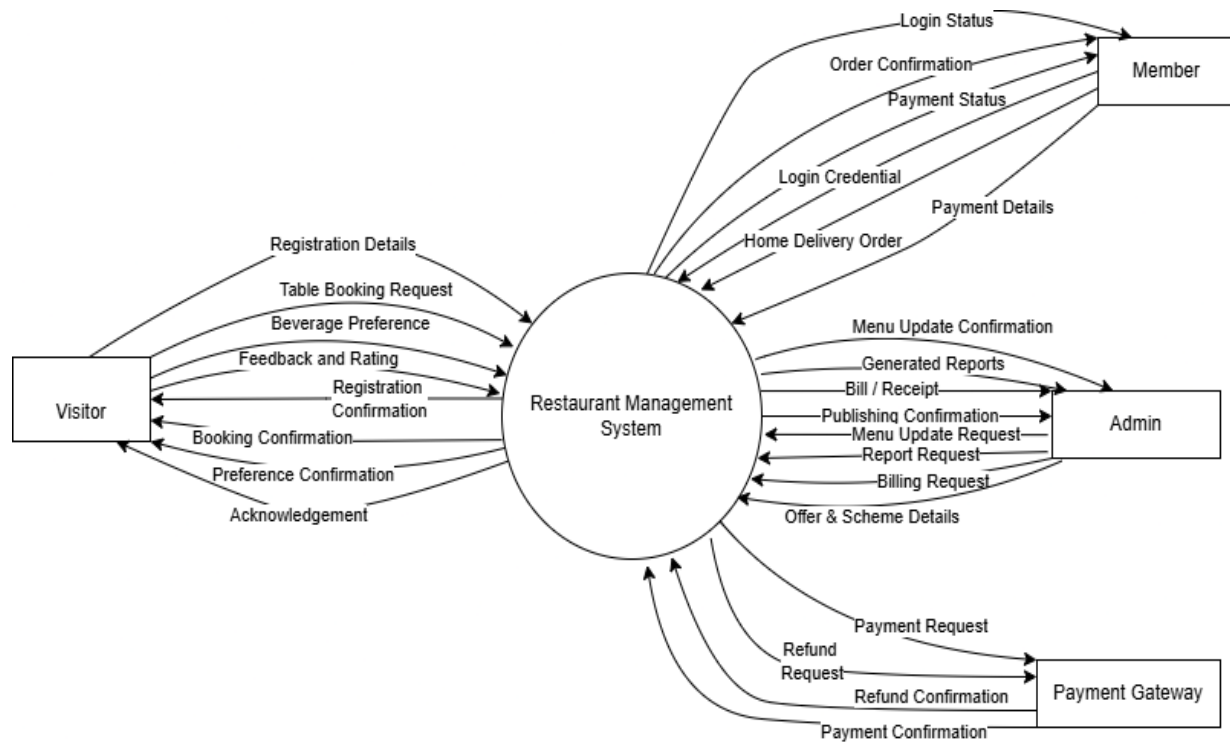


Figure 1 DFD Level 0

4.2 DFD Level 1 (Internal Model for ALL features)

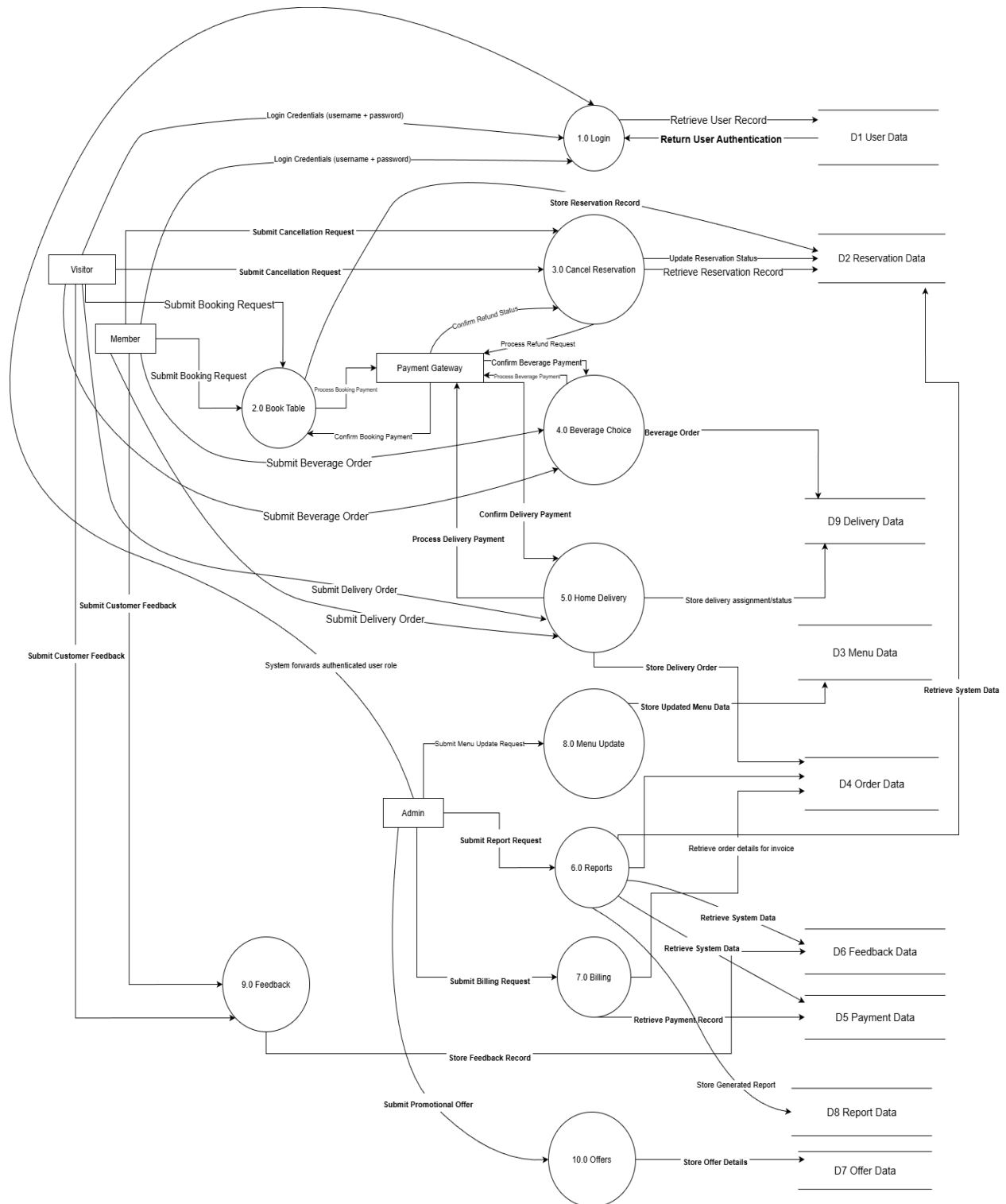
Processes (Level 1)

- P1 Manage Users (Register/Login/Logout)
- P2 Manage Table Booking
- P3 Manage Reservation Cancellation
- P4 Manage Beverage Choices
- P5 Manage Home Delivery Orders
- P6 Generate Reports & Prediction
- P7 Bill & Receipt Generation
- P8 Manage Menu (CRUD)
- P9 Manage Ratings & Feedback

- P10 Manage Offers & Schemes

Data Stores

- D1 Users
- D2 Tables
- D3 Reservations
- D4 Payments
- D5 MenuItem
- D6 DeliveryOrders
- D7 OrderItems
- D8 Bills/Receipts
- D9 Ratings/Feedback
- D10 Offers



4.3 DFD Level 2

4.3.1 Level 2 Process 1: P1 Manage Users (Login & Registration)

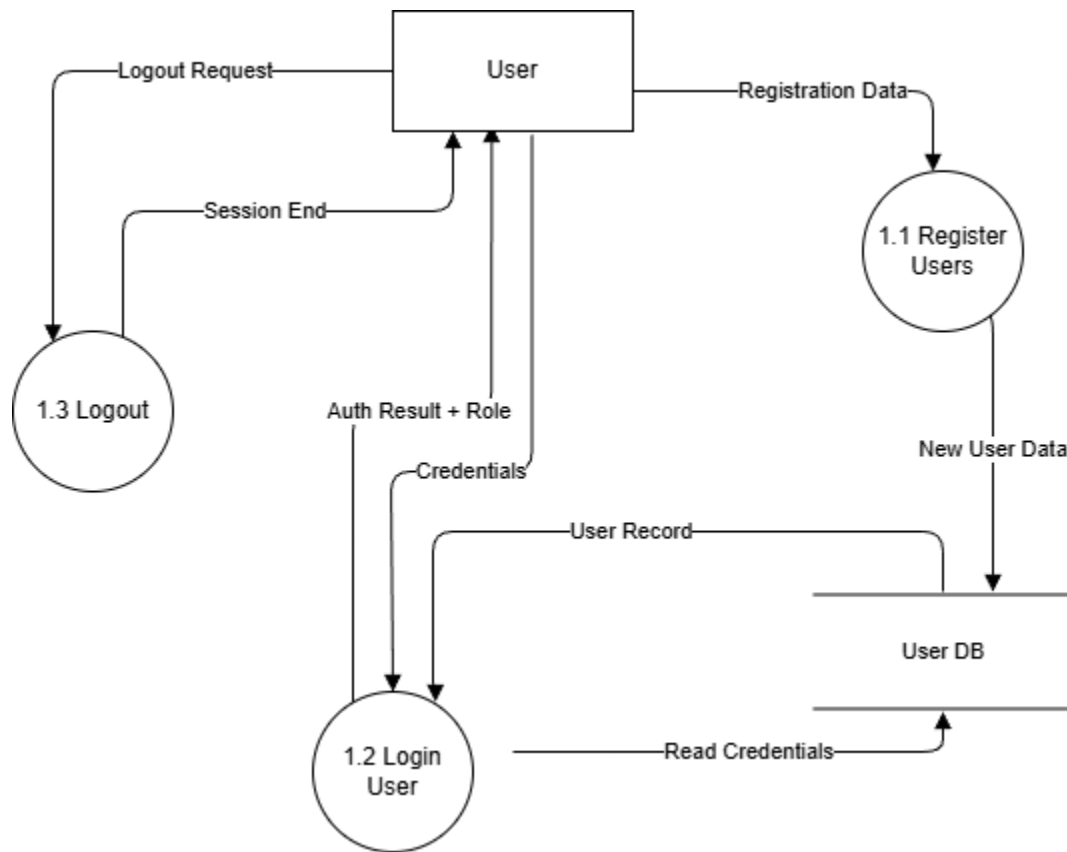


Figure 3 DFD Level 2: P1 Manage Users

4.3.2 Level 2 Process 2: P3 Manage Reservation Cancellation

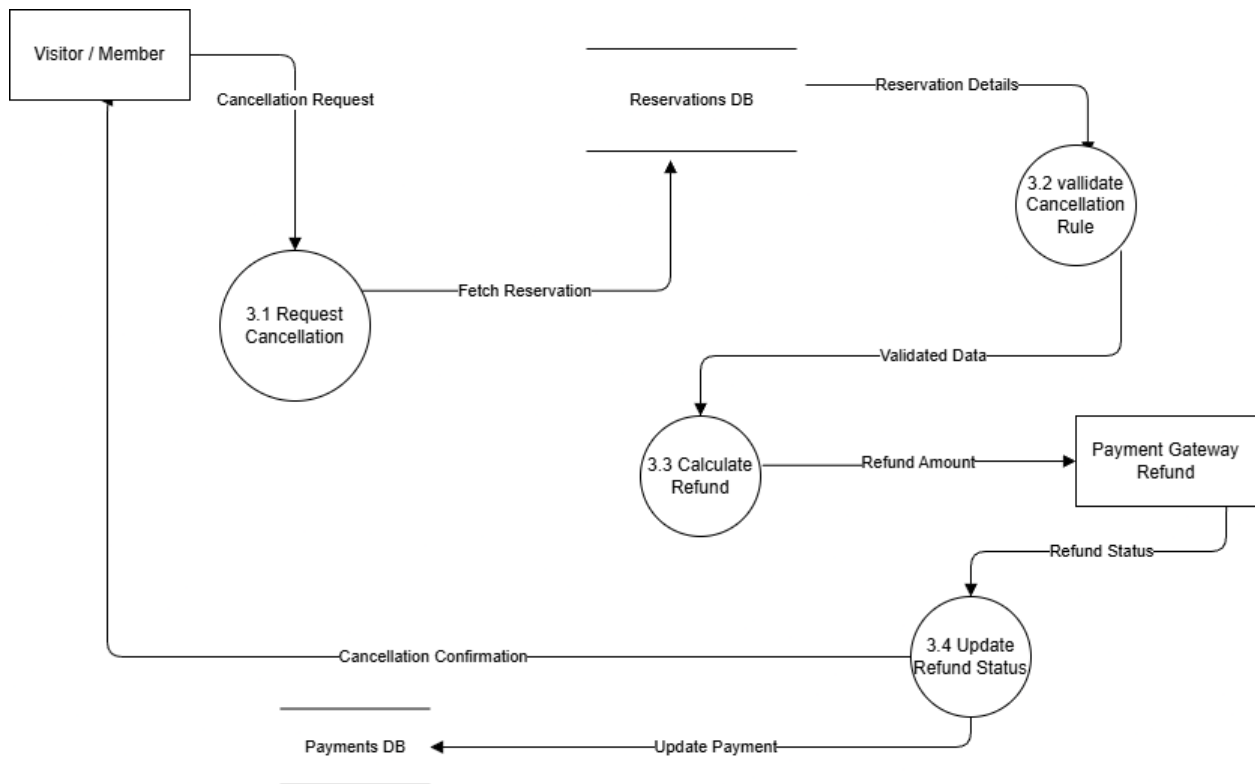
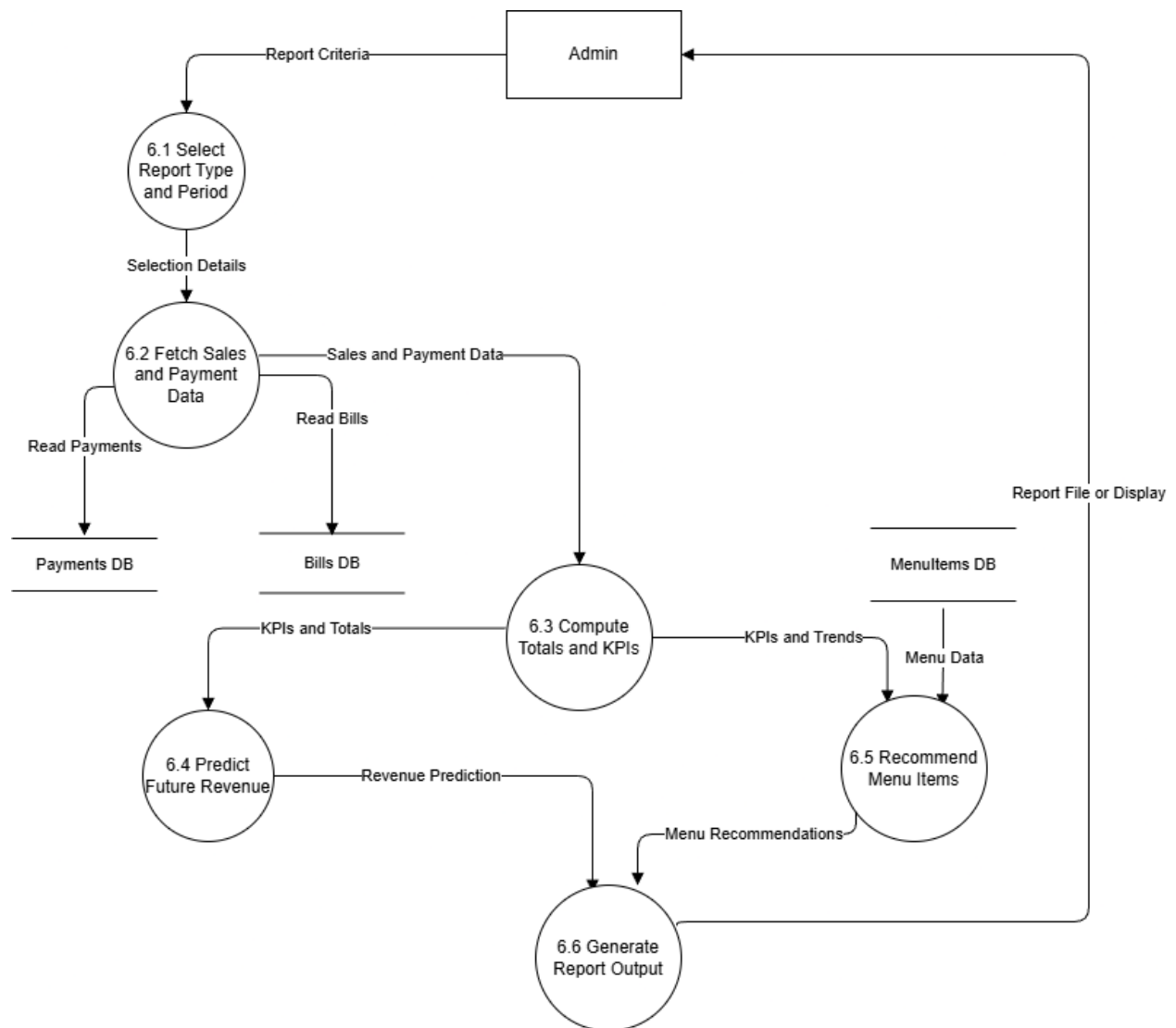


Figure 4 DFD Level 2: P3 Manage Reservation Cancellation

4.3.3 Level 2 Process 3: P6 Generate Reports & Prediction (Admin)*Figure 5 DFD Level 2: P6 Generate Reports and Prediction*

5 E-R diagram (Entity Relationship Diagram)

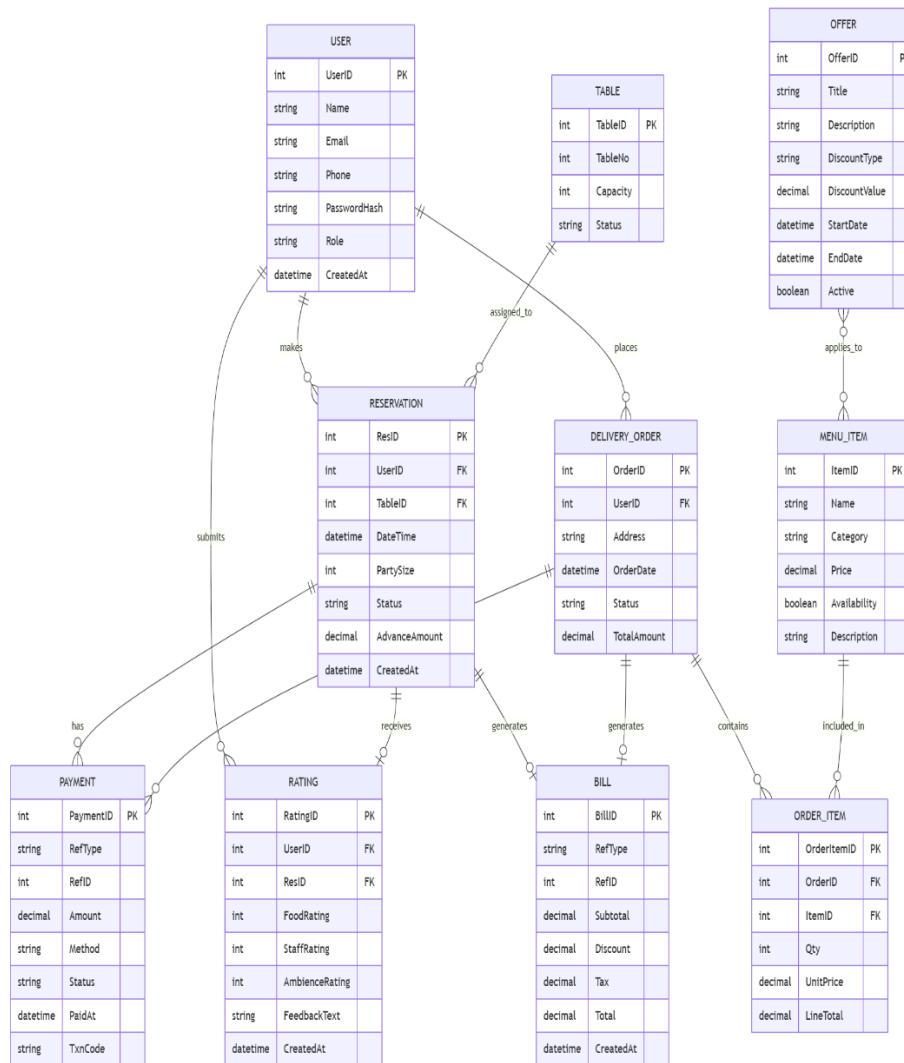


Figure 6 Entity Relationship Diagram

6 Data Dictionary

6.1 Data Stores

D1 Users

- Structure: (UserID, Name, Email, Phone, PasswordHash, Role, CreatedAt)

D2 Tables

- Structure: (TableID, TableNo, Capacity, Status)

D3 Reservations

- Structure: (ResID, UserID, TableID, DateTime, PartySize, Status, AdvanceAmount, CreatedAt)

D4 Payments

- Structure: (PaymentID, RefType, RefID, Amount, Method, Status, PaidAt, TxnCode)

D5 MenuItems

- Structure: (ItemID, Name, Category, Price, Availability, Description)

D6 DeliveryOrders

- Structure: (OrderID, UserID, Address, OrderDate, Status, TotalAmount)

D7 OrderItems

- Structure: (OrderItemID, OrderID, ItemID, Qty, UnitPrice, LineTotal)

D8 Bills/Receipts

- Structure: (BillID, RefType, RefID, Subtotal, Discount, Tax, Total, CreatedAt)

D9 Ratings/Feedback

- Structure: (RatingID, UserID, ResID, FoodRating, StaffRating, AmbienceRating, FeedbackText, CreatedAt)

D10 Offers

- Structure: (OfferID, Title, Description, DiscountType, DiscountValue, StartDate, EndDate, Active)

6.2 Major Data Flows**F1 Registration Data**

- Fields: Name, Email, Phone, Password, Role

F2 Login Credentials

- Fields: Email/Username, Password

F3 Table Availability Request

- Fields: DateTime, PartySize

F4 Booking Request

- Fields: UserID, TableID, DateTime, PartySize

F5 Payment Request

- Fields: RefType, RefID, Amount, PaymentMethod

F6 Payment Status

- Fields: RefType, RefID, Status, TxnCode

F7 Cancellation Request

- Fields: ResID, UserID, RequestDateTime

F8 Refund Details

- Fields: RefundAmount, DeductionAmount(0 or 500), RefundStatus

F9 Delivery Order Request

- Fields: UserID, Address, Items(ItemID, Qty)

F10 Bill/Receipt Output

- Fields: BillID, Total, Tax, Discount, Timestamp

F11 Report Parameters

- Fields: ReportType, StartDate, EndDate

F12 Report Output

- Fields: Totals, KPIs, Prediction, Recommendations

7 Process Specification

7.1 PSpec-1: Register User (P1.1)

Process ID: P1.1

Process Name: Register User

Description: This process is responsible for creating a new user account in the system. It allows a visitors or Members to provide registration details, validates the input information, ensures that the email address is unique, and stores the new user record securely in the system (Choudhary, 2024). Upon Successful registration, the system generates a unique user identifier and confirms the registration status to the user.

Inputs:

- Name
- Email
- Phone
- Password
- Role (visitor / Member)

Outputs:

- Registration status (Success / Failure)
- UserID

Data Stores:

- D1-Users

Process Logic:

1. Receive registration details from the user.
2. Validate required fields and verify the email formate.
3. Check D1 Users to determine whether the email already exists.
4. If the email exists, return a registration failure message.
5. Hash the user's password for security.
6. Store the new user record in D1-Users.
7. Generate a unique User ID.
8. Return a successful registration status with the generated Users ID.

7.2 PSpec-2: Login User (P1.2)

Process ID: P1.2

Process Name: Login User

Description: This process is responsible for authenticating an existing Admin or Member in the system. It verifies the user's login credentials against stored user records and, upon successful authentication, initiates a user session. The process ensures that only valid users are granted access based on their assigned role.

Inputs:

- Email
- Password

Outputs:

- Login status (Success / Failure)
- User Role
- Session token

Data Stores:

- D1-Users

Process Logic:

1. Receive login credentials from the user.
2. Retrieve the corresponding user record from DI Users using the Provided email.
3. If no matching user record is found, return an invalid login message.
4. Hash the entered password and compare it with the stored password hash.
5. If the password does not match, return a login failure message.
6. If credentials are valid, generate a session token.
7. Return a successful login status along with the user's role and session token.

7.3 PSpec-3: Logout User (P1.3)

Process ID: P1.3

Process Name: Logout User

Description: This process is responsible for securely terminating an active user session. It ensures that the session associated with the user is valid and then invalidates it by removing or deactivating the session information. This Prevents further access to the system using the same session credentials and confirms successful logout to the user.

Inputs:

- Session Token
- User ID

Outputs:

- Logout confirmation message

Data Stores:

- D1-Users (or Session Store)

Process Logic:

1. Receive a logout request containing the Session Token or User ID.
2. Validate the session against the active session data.
3. If the session is not found or is already invalid, return an “Invalid session” message.
4. Mark the user as logged out by removing or updating the session entry in D1-users (or the session store)
5. Destroy or invalidate the session token.
6. Return a logout success confirmation message to the user.

7.4 PSpec-4: Request Cancellation (P3.1)

Process ID: P3.1

Process Name: Request Cancellation

Description: This process handles cancellation requests submitted by a user for an existing reservation. It verifies whether the reservation belongs to the requesting user and ensures that the reservation exists in the system. Valid cancellation requests are then forwarded for further rule validation, such as cancellation timing and refund eligibility.

Inputs:

- Reservation ID (ResID)
- User ID
- Request Date

Outputs:

- Cancellation request status (Accepted / Rejected)

Data Stores:

- D3-Reservations

Process Logic:

1. Receive a cancellation request from the user.
2. Verify that the reservation exists and is associated with the given User ID.
3. If the reservation is not found or does not belong to the user, reject the cancellation request.
4. Forward the valid cancellation details for cancellation rule validation.

7.5 PSpec-5: Validate Cancellation Rule (P3.2)

Process ID: P3.2

Process Name: Validate Cancellation Rule

Description: This process is responsible for validating whether a reservation cancellation request complies with the defined business rules. It determines the type of cancellation by comparing the

reservation date with the cancellation request date and categorizes the request as either a same-day (late) cancellation or normal cancellation. The result of this validation is forwarded for refund determination.

Inputs:

- Reservation ID (ResID)
- Request Date

Outputs:

- Cancellation Rule result (Same day / normal Cancellation)

Data Stores:

- D3-Reservations

Process Logic:

1. Retrieve the reservation date associated with the given Reservation ID from D3 Reservations.
2. Compare the reservation date with the cancellation request date.
3. If the cancellation request is made is made on the same day as the reservation, classify it as a late cancellation.
4. Otherwise, classify it as a normal cancellation.
5. Forward the cancellation rule result to the refund calculation process.

7.6 PSpec-6: Calculate Refund (P3.3)

Process ID: P3.3

Process Name: Calculate Refund

Description: This process is responsible for calculating the refund amount for a cancelled reservation based on the validate cancellation rule. It determines whether a deduction should be applied and computes the final refund amount accordingly before forwarding the result for

refund processing.

Inputs:

- Advance Amount
- Cancellation Rule Result

Outputs:

- Refund Amount
- Deduction Amount

Data Stores:

D3-Reservations

Process Logic:

1. Check the cancellation rule result received from the validation process.
2. If the cancellation is classified as a same-day (late) cancellation, calculate the refund by deducing the predefined amount from the advance payment.
3. If the cancellation is classified as a normal cancellation, set the refund amount equal to the advance payment.
4. Determine and record the deduction amount based on the applied rule.
5. Forward the calculated refund and deduction amounts for refund update and processing.

7.7 PSpec-7: Update Refund Status (P3.4)

Process ID: P3.4

Process Name: Update Refund Status

Description: This process is responsible for finalizing the refund operation and updating the reservation status after a cancellation. It records the refund transaction, coordinates with the payment system to process the refund, updates payment and reservation records, and informs the user about the outcome of the refund request.

Inputs:

- Reservation ID (ResID)
- Refund Amount

Outputs:

- Refund status
- Cancellation confirmation Message

Data Stores:

- D3-Reservations
- D4-Payments

Process Logic:

1. Create a refund transaction record in D4 Payments.
2. Send the refund request to the payment gateway for processing.
3. Update the payment status based on the response received from the payment gateway.
4. Update the reservation status in D3 Reservation to “Cancelled”.
5. Notify the User with the refund result and cancellation confirmation.

7.8 PSpec-8: Select Report Type & Period (P6.1)**Process ID:** P6.1**Process Name:** Select Report Parameters

Description: This process is responsible for receiving and validating report selection criteria provided by the Administrator. It verifies the administrator’s authorization, validates the selected report type and date range, and ensures that the input parameters are logically correct before forwarding them to the data retrieval process for report generation.

Inputs:

- Report Type (Daily / Weekly / Monthly / Custom)
- Start Date

- End Date
- Admin ID

Outputs:

- Validated report parameters
- Error Message (if validation fails)

Data Stores:

- D1-Users
- D4-Payments

Process Logic:

1. Receive report parameters from the Administrator.
2. Verify the Admin ID and confirm the user's role using D1 Users.
3. Validate the selected report type.
4. Validate that the start date is less than or equal to the end date and that the selected dates are not in the future.
5. Optionally check the existence of relevant data within **D4 Payments** for the selected date range.
6. If any validation fails, return an appropriate error message.
7. Forward the validated report parameters to the next process for report data retrieval.

7.9 PSpec-9: Fetch Sales / Payment Data (P6.2)

Process ID: P6.2

Process Name: Fetch Payment Data

Description: This process is responsible for retrieving financial and transactional data required for report generation. It collects payment records, refund transactions, and billing information within the specified date range and prepares a consolidated dataset to be used for further

computation and analysis.

Inputs:

- Date range

Outputs:

- Payment dataset

Data Stores:

- D4-Payments
- D8-Bills

Process Logic:

1. Query successful payment transaction from D4 Payments within the specified date range.
2. Retrieve refund transaction records associated with the same period.
3. Fetch corresponding billing records from D8 Bills.
4. Combine and forward the retrieved data as a dataset for report computation and generation.

7.10 PSpec-10: Compute Totals & KPIs (P6.3)

Process ID: P6.3**Process Name:** Compute KPIs

Description: This process is responsible for computing financial totals and key performance indicators (KPIs) required for management reports. Using the validated report parameters and the retrieved payment dataset, the process calculates overall sales, refunds, net revenue, and other relevant metrics that help assess business performance over a selected period.

Inputs:

- Report parameters
- Payment dataset (within selected date range)

Outputs:

- Total Sales
- Total Refunds
- Net Revenue,
- Computed KPIs

Data Stores:

- D4-Payments
- D8-Bills / Receipts

Process Logic:

1. Retrieve successful payment records from **D4 Payments** for the specified reporting period.
2. Retrieve refund transactions records from **D4 Payments** within the same period.
3. Optionally retrieve billing totals from **D8 Bills / Receipts** for validation or cross-checking.
4. Calculate Total Sales by summing all successful payment amounts.
5. Calculate Total Refunds by summing all refund amounts.
6. Compute **NetRevenue = TotalSales – TotalRefunds**.
7. Calculate Key performance indicators such as:
 - Average revenue per day,
 - Number of transactions,
 - Growth rate compared to previous period.
8. Forward computed totals and KPIs to the next process for report generation or presentation.

7.11 PSpec-11: Predict Future Revenue (P6.4)**Process ID:** P6.4**Process Name:** Predict Revenue**Description:** This process is responsible for estimating future revenue based on historical

financial data. It analyzes past net revenue trends to generate predicted revenue values, which can be used by management for planning and decision-making purpose. The predication provides an approximate outlook rather than an exact forecast.

Inputs:

- Historical net revenue

Outputs:

- Predicted revenue values (Monthly / Yearly)

Data Stores:

- D4-Payments

Process Logic:

1. Retrieve historical net revenue data from D4 Payments.
2. Calculate the average monthly revenue over the selected historical period.
3. Apply a simple trend analysis or moving average technique to identify revenue patterns.
4. Estimate predicated revenue values for the next month and the upcoming year.
5. Forward the predication results to the report generation or presentation process.

7.12 PSpec-12: Recommend Menu Items (P6.5)**Process ID: P6.5****Process Name: Recommend Menu Items**

Description: This process is responsible for generating menu item recommendations based on historical order data. It analyzes previously place orders to identify top-selling items and produces a list of frequently ordered menu items. These recommendations help management understand customers preferences and support promotional or menu planning decisions.

Inputs:

- Order History

Outputs:

- Recommended Menu Item List

Data Stores:

- D6-Delivery Orders
- D7-Order Items

Process Logic:

1. Retrieve order and item details from D6 Delivery Orders and D7 Orders Items.
2. Calculate the total based on quantity sold in descending order.
3. Sort menu items based on quantity sold in descending order.
4. Select the top N items as recommended menu items.
5. Generate and forward the recommended item list to the reporting or presentation process.

7.13 PSpec-13: Generate Report Output (P6.6)**Process ID: P6.6****Process Name: Generate Report Output**

Description: This process is responsible for preparing and presenting and presenting the final report output for the Administrator. It combines computed totals, key performance indicators, revenue predictions, and menu recommendations into a structured and readable format. The generated report can be displayed on-screen or exported for further use.

Inputs:

- Computed Totals
- Key Performance Indicators (KPIs)
- Revenue Predictions
- Recommended Menu Items

Outputs:

- Final Report Output (Screen View / PDF)

Data Stores:

- D8-Bills

Process Logic:

1. Format the computed totals and KPIs into a structure report layout.
2. Incorporate revenue predication values into the report.
3. Include recommended menu items and supporting summaries.
4. Generate the final report output in a suitable display or export format.
5. Present the report to the Administrator through screen view or export it as a file.

8 Design specification:

9 Progress logs:

Phase	Activities Performed	Deliverables Completed	Challenges Faced	Resolution and Improvements
Phase 1- Requirement Study	Carefully read and analyzed the case study. Identified key features such as login/registration, table booking, beverage selection, cancellation, and delivery orders. Defined the scope of the system and user roles.	Drafted the Business Case and initial SRS outline.	Difficulty understanding some business rules.	Clarified assumptions and structured the requirements logically.
Phase 2 – System Modeling (Level 0)	Designed the Context Level Diagram showing system interaction with Visitor, Member, Admin, Payment Gateway, and Delivery Service.	Completed Level 0 DFD.	Confusion about balancing external entities.	Corrected flows so each external input/output matches the system boundary.
Phase 3 – DFD Level 1 Design	Broke down the system into major processes including authentication, table booking, payment processing, order management, menu update, offers, and rating.	Completed DFD Level 1 for all features.	Missing data stores and unclear flows.	Added required data stores and refined the data flow connections.
Phase 4 – DFD Level 2 Refinement	Further decomposed three internal processes (excluding individual features). Ensured logical and balanced flow between sub-processes and databases.	Completed DFD Level 2 diagrams.	Too many process links caused clutter.	Simplified and reorganized the diagram for clarity.
Phase 5 – ERD Development	Created the Entity Relationship Diagram based on the system requirements. Defined entities such as User, Reservation, Order, Payment, Menu, Rating, and Offers.	Completed ERD with correct cardinalities.	Incorrect modality and relationships.	Reviewed the ERD and updated relationships to meet the business rules.

Phase 6 – Data Dictionary Creation	Documented all major data flows, data stores, and entity definitions. Included attribute names, descriptions, and data types.	Completed full Data Dictionary.	Difficulty maintaining consistency between diagrams and dictionary.	Cross-checked each flow and entity with the ERD and DFD.
Phase 7 – Process Specifications	Wrote detailed logic for elementary processes using structured English. Included validation steps and decision rules.	Completed Process Specifications.	Some process descriptions were too vague.	Expanded them with input/output details and validation logic.
Phase 8 – Design Specification	Derived the Structure Chart from the DFD diagrams. Designed the hierarchical module breakdown for the “Book Table” feature.	Completed Structure Chart.	Arrow types and module flow not correct initially.	Re-designed the chart using standard data/control couples.
Phase 9 – Final Report Compilation	Organized and formatted the entire report according to coursework guidelines. Performed proofreading and quality checks.	Completed final submission draft.	Ensuring all parts aligned correctly.	Verified consistency between all diagrams and sections.

10 Individual Task:

10.1 Member 1:

Student Name: Pappu Kumar Singh Kushwaha

LMU ID: 24058814

Function chosen: Book Table

10.2 Context Level Diagram:

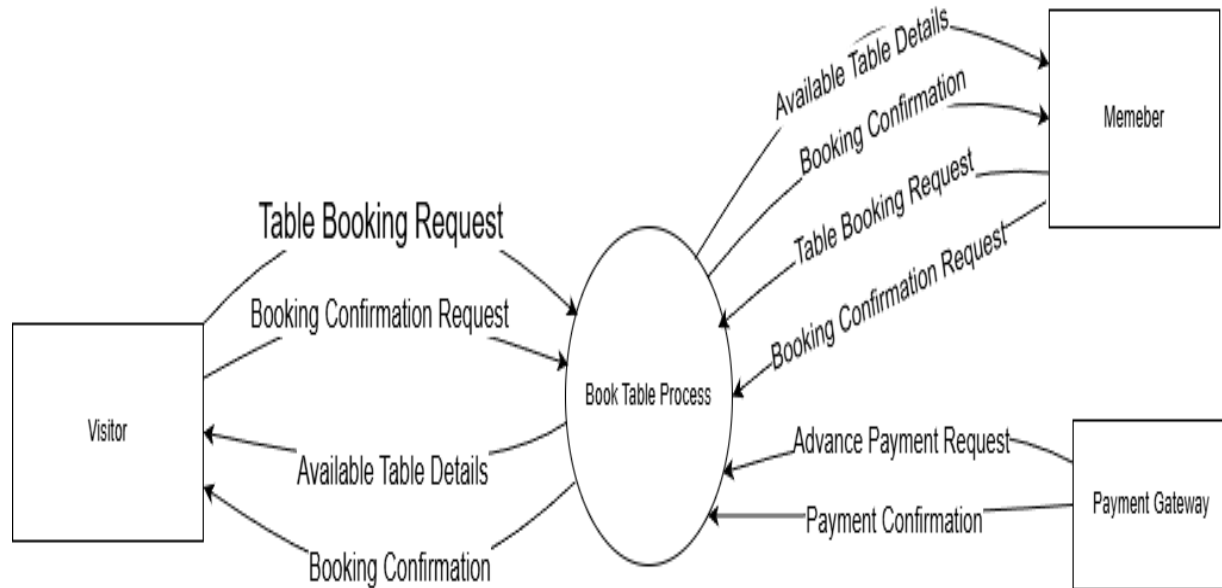


Figure 7 DFD Book Table Level 0

10.3 DFD Level-1:

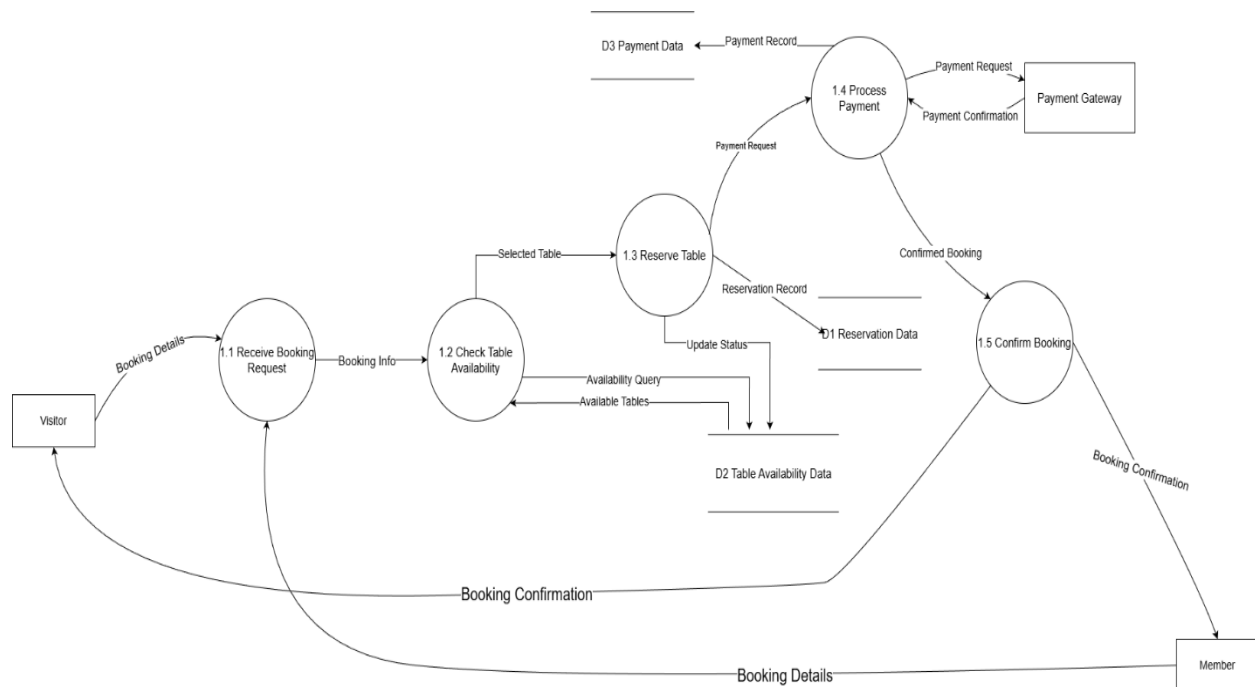
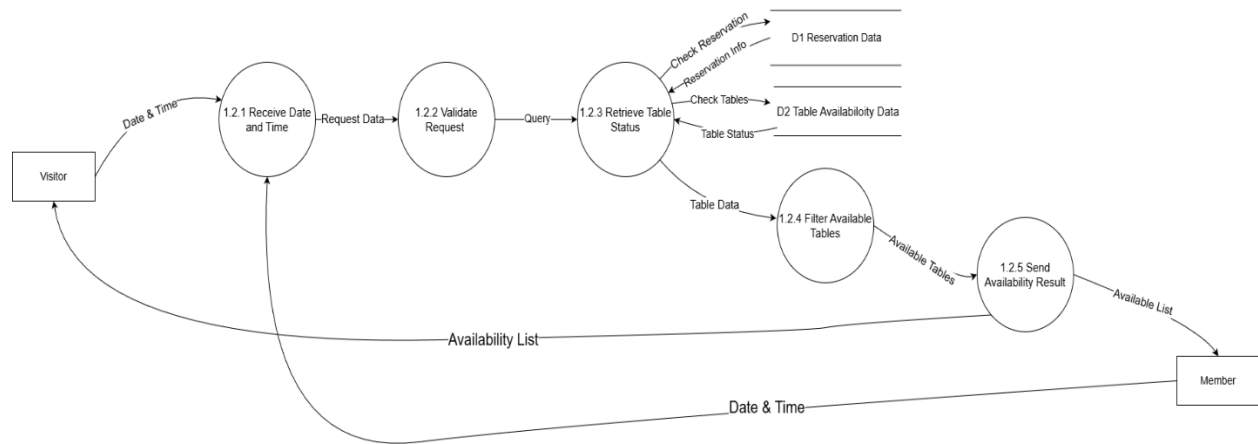


Figure 8 DFD Book Table Level 1

10.4 DFD Level-2 (For Specification):*Figure 9 DFD Book Table Level 2*

10.5 Design Specification:

Structure Chart for the Book Table Function.

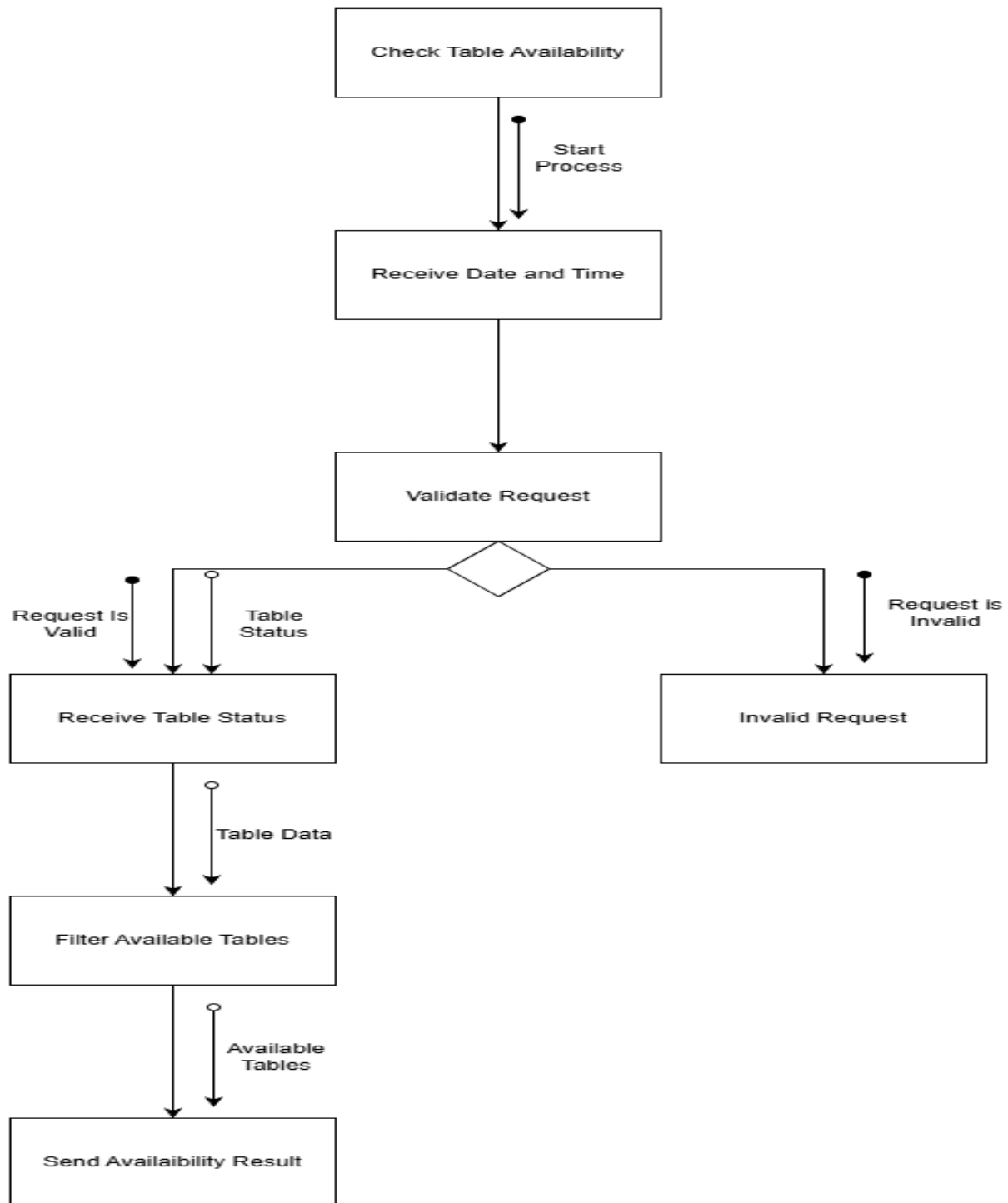


Figure 10: Book Table Structure Chart

Module specification (MSpecs):

MODULE NAME: Book Table

PURPOSE: The Purpose of the Book Table module is to allow Visitors and Members to reserve a table at the restaurant for a selected date and time. This module handles booking requests. Checks table availability, reserves the selected table processes advance payment, and confirms the booking to the users. The module ensures that reservations are recorded accurately and prevents double booking of tables.

PSEUDOCODE:

START Book Table Module

RECEIVE booking request from visitor or Member

CAPTURE booking details (date, time, number of guests)

VALIDATE booking details

IF booking details are invalid

RETURN error message

END IF

CHECK table availability for selected date and time

IF table is not available

RETURN availability message

END IF

RESERVE selected table

STORE reservation details

GENERATE payment request

SEND payment request to Payment Gateway

RECEIVE payment confirmation

IF payment successful

STORE payment record

CONFIRM booking

SEND booking confirmation to user

ELSE

CANCEL reservation

SEND payment failure message

END IF

END Book Tabel Module

INPUT PARAMETERS:

- Booking Date
- Booking Time
- Number of Guests
- Customer Type (Visitor / Member)
- Table Selection
- Payment Details

OUTPUT PARAMETERS:

- Table Availability Details
- Booking confirmation Message
- Booking Reference Number
- Payment Status

GLOBAL VARIABLES:

- Total Number of Tables
- Restaurant Operating Hours
- Advance Payment Percentage

LOCAL VARIABLES:

- Selected Table ID
- Reservation ID
- Payment Transaction ID
- Booking Status

CALLS:

- Check Table Availability Module
- Reserve Table Module

- Process Payment Module
- Confirm Booking Module
- External Library / Services: Payment Gateway API

CALLED BY:

- Visitor Interface
- Member Interface
- Restaurant Management System (Main Controller)

10.6 Member 2:

Student Name: Priyans Bhatt

LMU ID: 21049638

Function Chosen: Rate the Experience

Context Level Diagram:

Description:

- ☐ Level-0 DFD shows the overall design of the Rate the Experience module and shows high-level view of how major entities interact with the system but does not go into detail on how internal processing occurs.
- ☐ There are two external entities for the Rate the Experience module: Visitors and members. Both entities can provide feedback, comments and ratings to the Rate the Experience central process.
- ☐ Rate the Experience collects this information and stores it in the Feedback Data Store (D1 -Feedback Data) for future reference and analysis.
- ☐ When the system successfully completes processing of the submitted feedback/ratings, the system sends back:
 - An Acknowledgement message to inform that the system received the rating/feedback.
 - A Submission Confirmation to inform the user that the feedback/rating has been successfully recorded in the system.

□ In this way the Level-0 DFD provides a high-level overview of the interactions between feedback users and the feedback system. The Level-0 DFD serves as the starting point to develop more detailed Level-1 and Level-2 DFD diagrams.

Context Level Diagram:

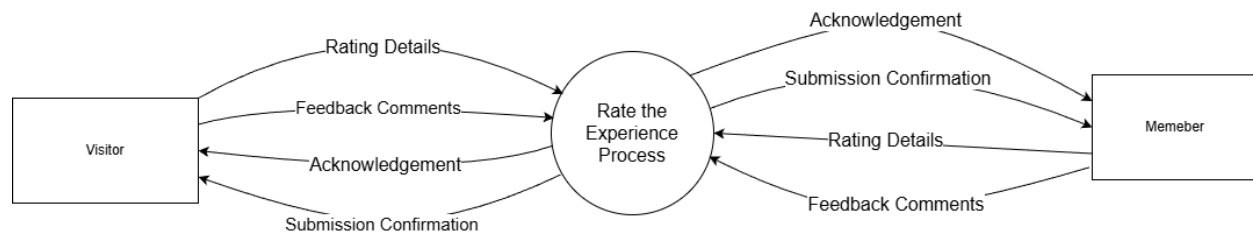


Figure 11 Rate the Experience DFD Level 0

DFD LEVEL-1:

Description:

- The Level 1 Data Flow Diagram shows how a Restaurant Service’s Rate the Experience module functions internally.
- The Rate the Experience module allows both visitors and members to rate and leave comments on the quality of their dining experience.
- The first interaction with the module occurs when the user submits their overall rating to the Receive Rating (2.1) process. The system then uses that rating to verify its accuracy via the Validate Rating (2.2) process.
- Once validated, the rating is stored within the Store Feedback (2.4) process. At the same time as submitting a rating, users may also submit written comments via the Receive Feedback (2.3) process, and both submissions get passed to the same process for storage.
- The storage process saves all submitted rating and comment information into D1 Feedback Data. Then, after the information has been saved, the storage process sends a confirmation to the Send Acknowledgement (2.5) process, and the Send Acknowledgement process generates and sends an acknowledgement message back to the visitor or member.

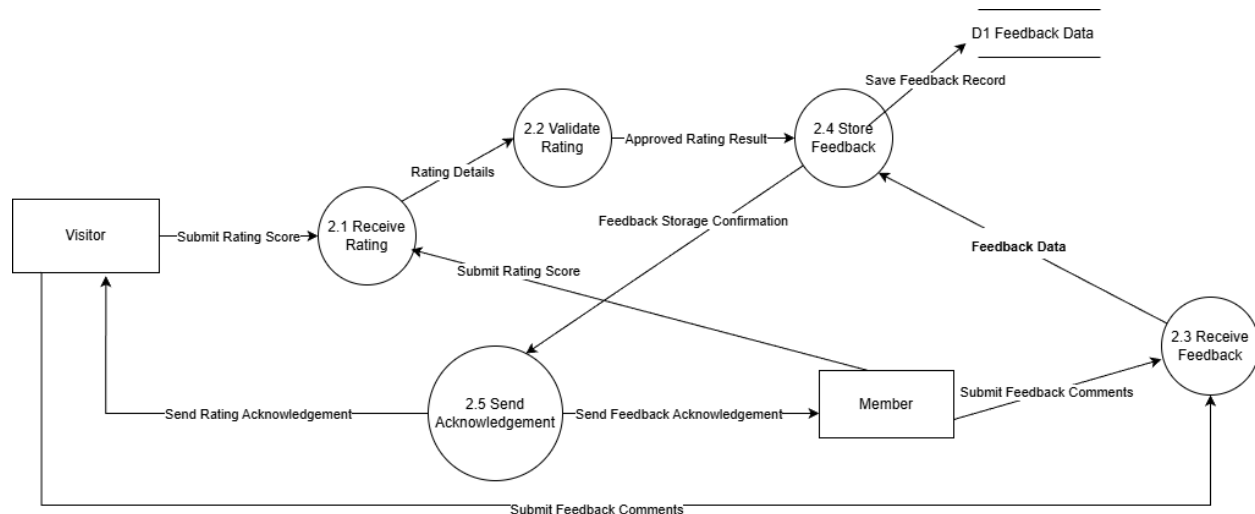


Figure 12 Rate the Experience DFD Level 01

DFD LEVEL-2 (For Specification):

Description:

- The Level-2 Data Flow Diagram expands the Level-1 Data Flow Diagram in the section 2.2 Validate Rating. At Level-1 the section identifies the process for validating the rating but does not explain in detail the method of verifying the rating.
- Level-2 breaks down the actual steps that are taken to validate the rating into logical groups.
- The first step in validating the rating is the 2.2.1 Check Rating Range, which checks the Submitted Rating to see if it is within the range that is acceptable.
- The second step in the validation process is 2.2.2 Validate Rating Format, which checks to ensure that the Submitted Rating has been entered in the correct numeric format.
- Once the Submitted Rating has been validated through these steps and meets all of the required validation criteria, it will be approved through the 2.2.3 Approve Valid Rating process. After being approved, the Submitted Rating will be stored

in the D1 Feedback Data Store as a validated rating.

- If the Submitted Rating does not meet either the acceptable value range or the correct number format, it will be assessed using the 2.2.4 Reject Invalid Rating process. This process will inform the user of the validation error and require that the user corrects their Submitted Rating before proceeding.
- Level-2 provides a clear view of both validating the rating, understanding how it is approved and stored, as well as understanding how to reject the rating and notify the user of the issues associated with their submitted rating.

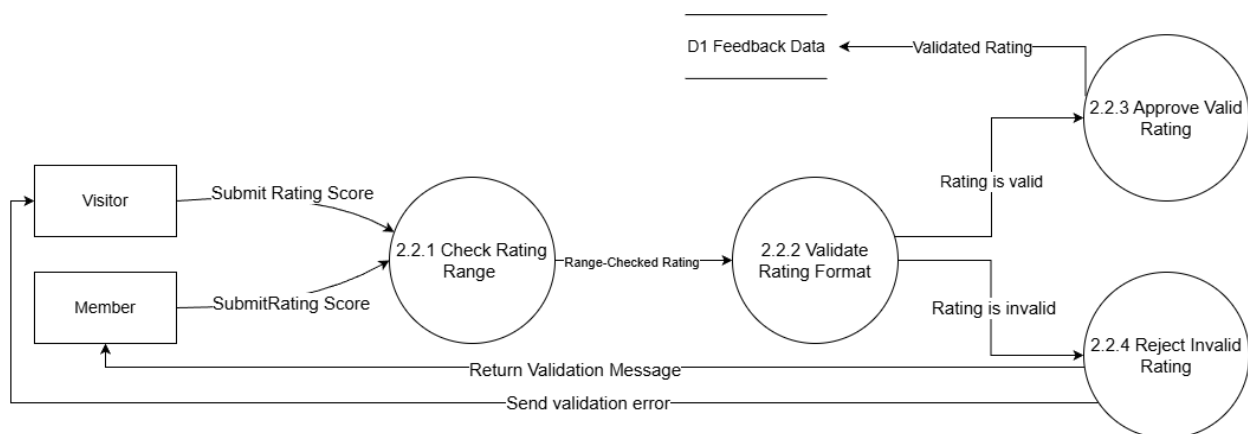


Figure 13 Rate the Experience DFD Level 2

Design Specification:

Structure Chart for the Particular Function:

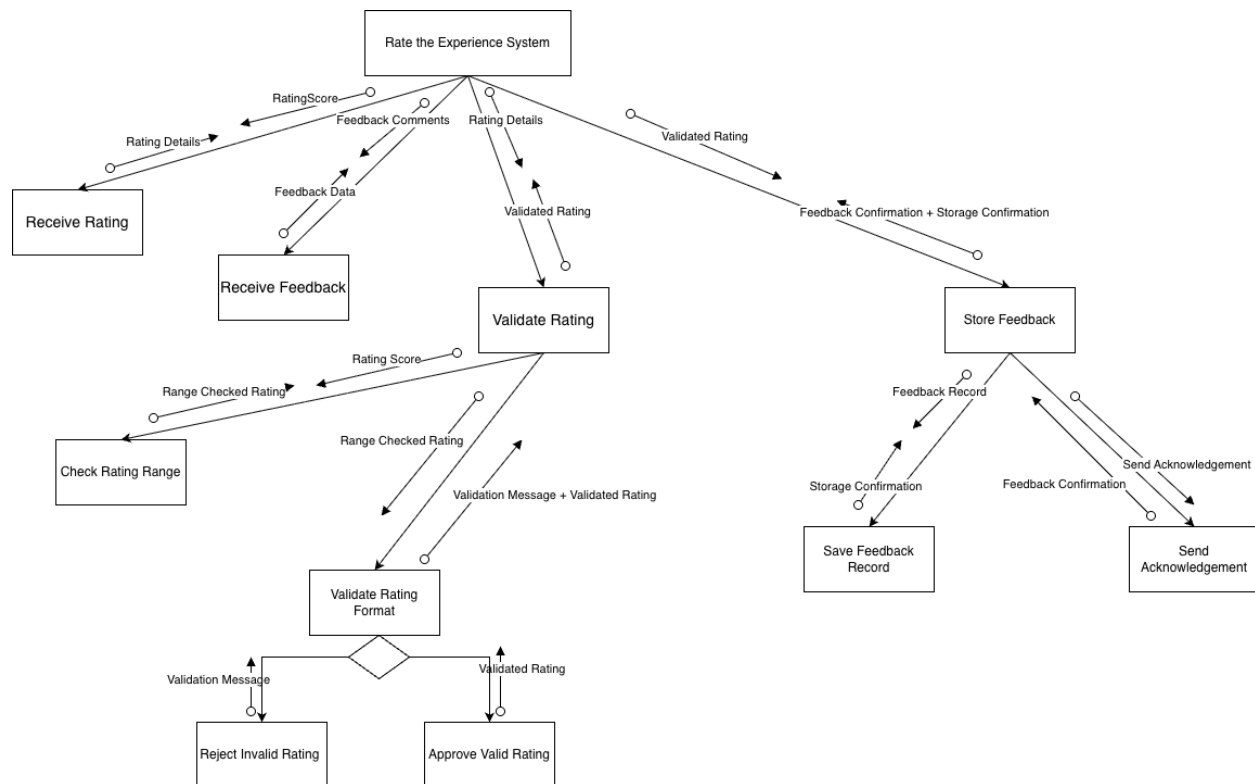


Figure 14: Structured chart of Rate experience

Module Specifications (MSpecs):

MODULE NAME: Rate The Experience Module

PURPOSE: The purpose of the Rate the Experience module is to allow Visitors and Members to submit ratings and feedback regarding their experience with the restaurant. This module collects user ratings, validates the input, stores Feedback information, and provides acknowledgement to the user. The module helps the restaurant evaluate service quality and improve customer satisfaction.

PSEUDOCODE:

START Rate the Experience Module

Receive rating score from Visitor or Member

Receive feedback comments from Visitor or Member

VALIDATE rating score

IF rating score is invalid

RETURN error message

END IF

VALIDATE feedback comments

IF feedback comments are empty

SET feedback as optional

END IF

STORE rating and feedback details in feedback data store

GENERATE acknowledgement message

SEND acknowledgement to user

END Rate the Experience Module

INPUT PARAMETERS:

- Rating Score
- Feedback Comments
- User Type (Visitor / Member)

OUTPUT PARAMETERS:

- Feedback Submission Confirmation
- Acknowledgement Message

GLOBAL VARIABLES:

- Maximum Rating Value
- Minimum Rating Value

LOCAL VARIABLES:

- Feedback ID
- Validation Status
- Submission Timestamp

CALLS:

- Validate Rating Module
- Store Feedback Module
- External Library / Components: Input Validation Utility

CALLED BY:

- Visitor Interface

- Member Interface
- Restaurant Management System (Main Controller)

10.7 Member 3:

Student Name: Pappu Yadav

LMU ID: 24058835

Function Chosen: Set Beverage Choices

Context Level Diagram:

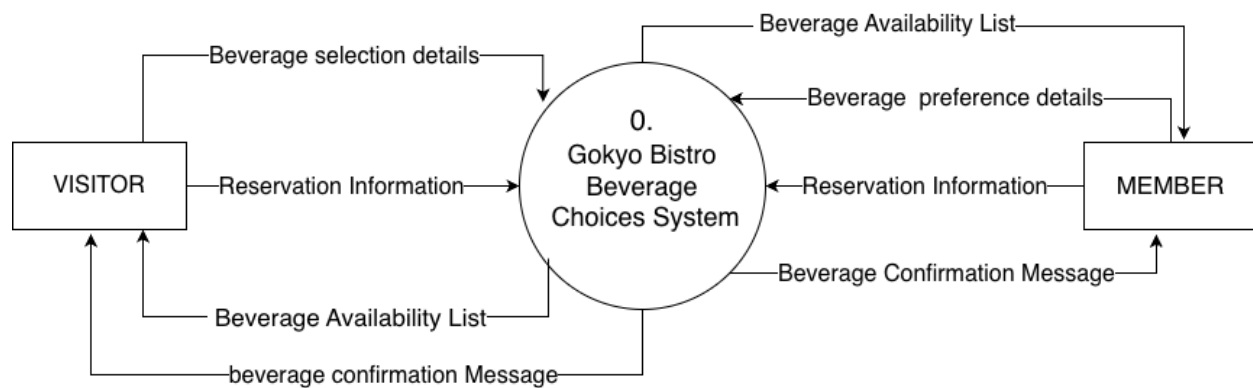


Figure 15 Set Beverage DFD Level 0

DFD LEVEL-1:

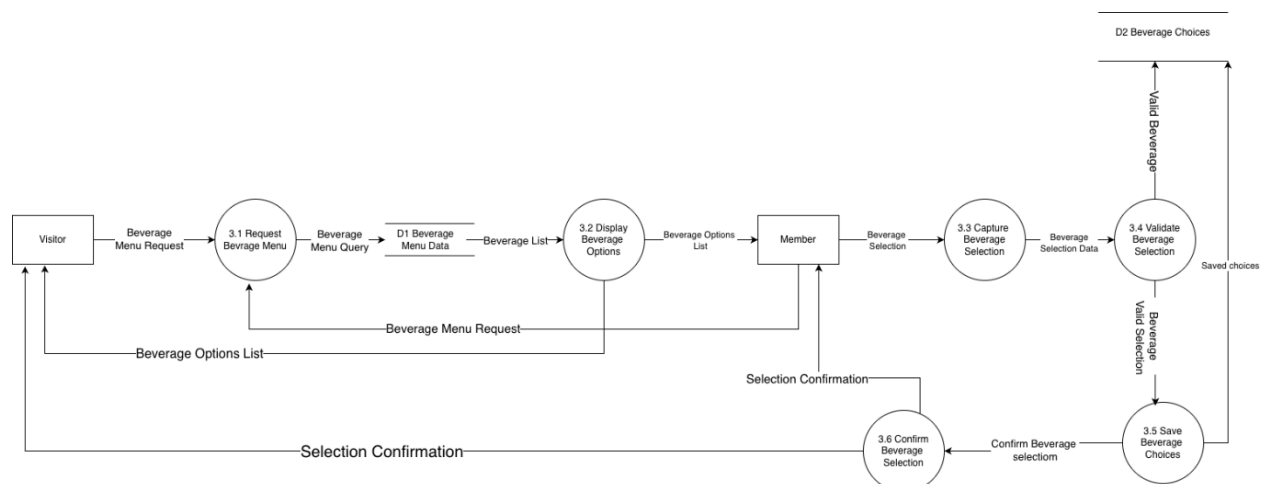
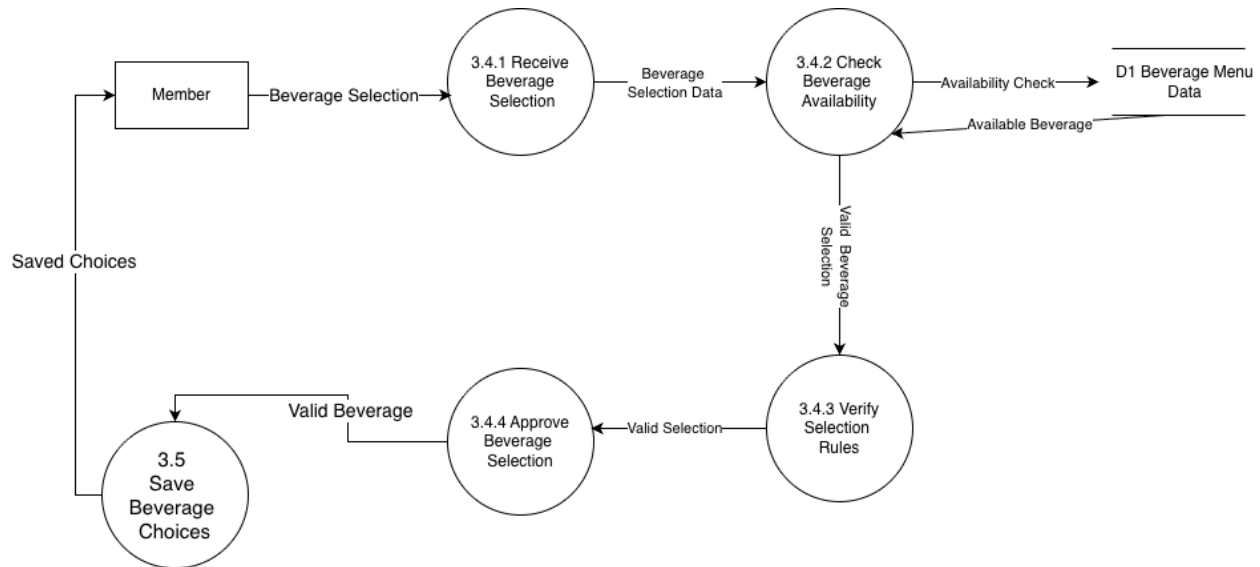
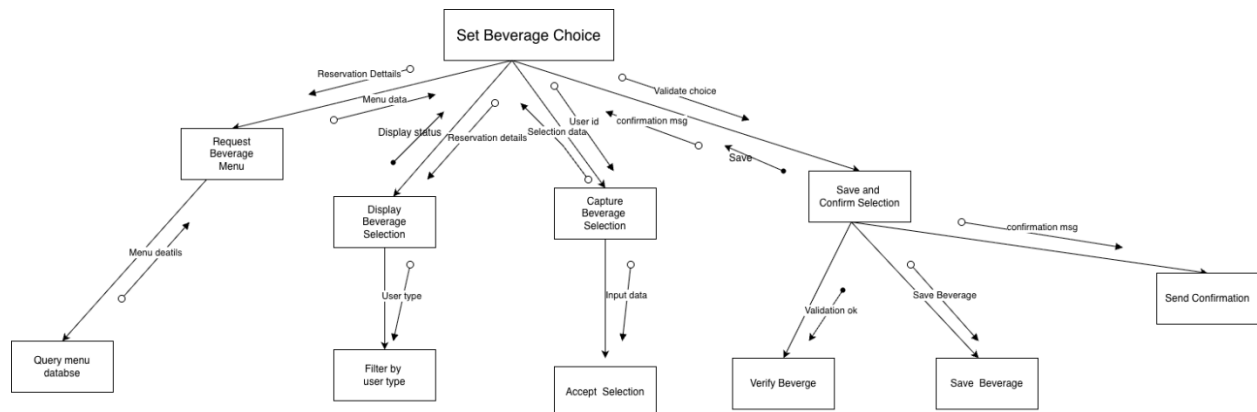


Figure 16 Set Beverage DFD Level 1

DFD LEVEL-2 (For Specification):*Figure 17 Set Beverage DFD Level 2***Design Specification:****Structure Chart for the Particular Function:***Figure 18: Structure chart of Set Beverage***Module Specifications (MSpecs):****MODULE NAME:** Set Beverage Choices Module

PURPOSE: The purpose of the Set Beverage Choices module is to allow Visitors and Members to select preferred beverages while making a reservation or order. This module retrieves available beverage options, captures user selections, validates the selected beverages based on availability and rules, stores the choices, and confirms the selection to the user. The module ensures that only valid and available beverages are selected.

PSEUDOCODE:

START Set Beverage Choices Module

RECEIVE beverage menu request from Visitors or Member

RETRIVE available beverage list

DISPLAY beverage options to user

RECEIVE selected beverages from user

VALIDATE selected beverages

IF any selected beverage is unavailable

RETURN error message

END IF

IF selection violates selection rules

RETURN validation error

END IF

STORE validated beverage choices

GENERATE confirmation message

SEND confirmation to user

END Set Beverage Choices Module

INPUT PARAMETERS:

- Beverage Menu Request
- Selected Beverages
- User Type (Visitor / Member)

OUTPUT PARAMETERS:

- Available Beverage List
- Beverage Selection Confirmation Message

GLOBAL VARIABLES:

- Maximum Beverage Selection Limit
- Beverage Availability Status

LOCAL VARIABLES:

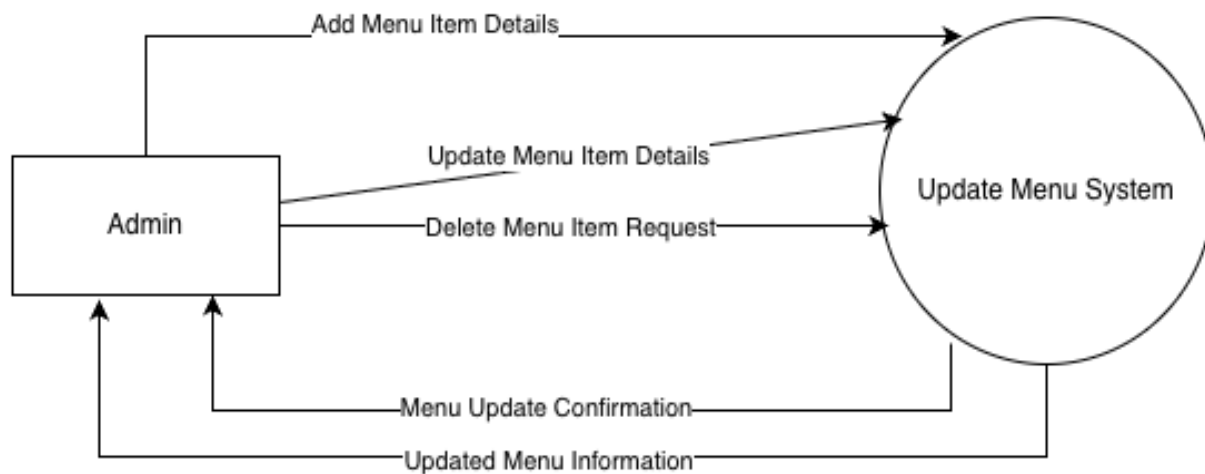
- Selected Beverage IDs
- Validation Status
- Selection Timestamp

CALLS:

- Retrieve Beverage Menu Module
- Validate Beverage Selection Module
- Store Beverage Choices Module
- External Component / Library: Menu Data Access Service

CALLED BY:

- Visitor Interface
- Member Interface
- Reservation / Order Management Controller

10.8 Member 4:**Student Name: Bishal Yadav****LMU ID: 24058807****Function Chosen: Update Menu****Context Level Diagram:***Figure 19 Update Menu DFD Level 0***DFD LEVEL-1:**

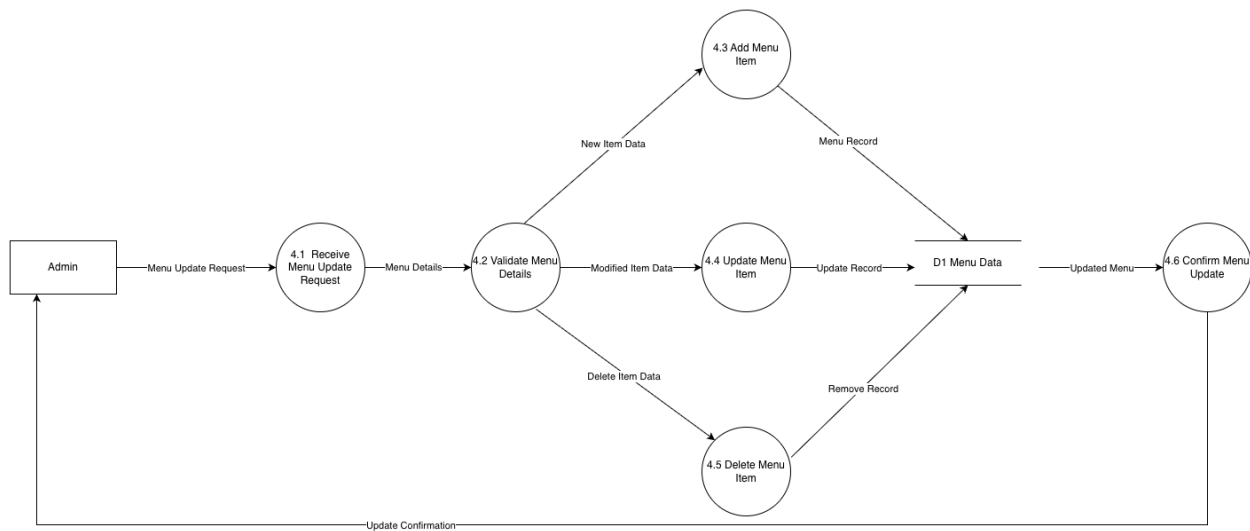


Figure 20 Update Menu DFD Level 1

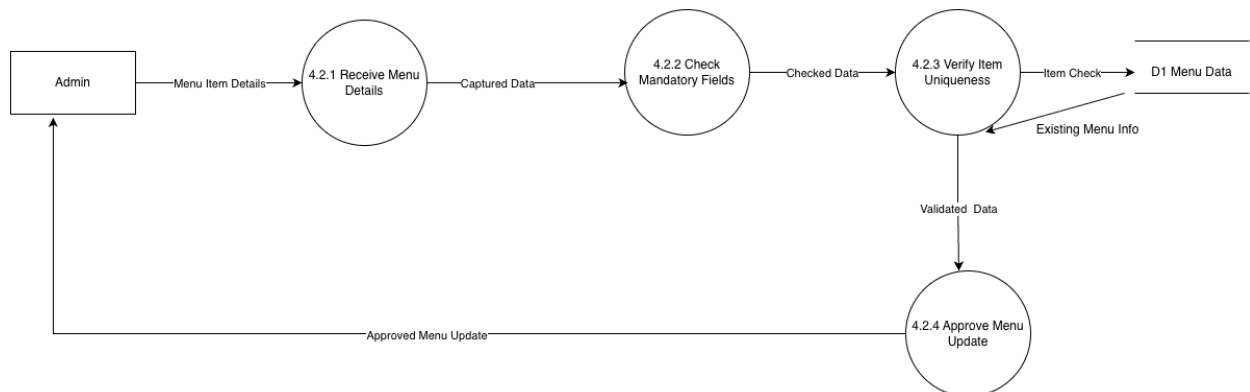
DFD LEVEL-2 (For Specification):

Figure 21 Update Menu DFD Level 2

Design Specification:**Structure Chart for the Particular Function:****Module Specifications (MSpecs):****MODULE NAME:** Update Menu Module

PURPOSE: The purpose of the Update Meny module is to allow the Administrator to manage the restaurant's menu by adding new items, updating existing items, or removing items that are no longer available. This module ensures that menu information remains accurate, up to date, and consistent for users interacting with the system.

PSEUDOCODE:

STRT Update Menu Module

RECEIVE menu update request from Admin

CAPTURE menu item details

VALIDATE menu details

IF menu details are invalid

RETURN validation error message

END IF

IF request type is **ADD**

ADD new menu item

ELSE IF request type is **UPDATE**

UPDATE existing menu item

ELSE IF request type is **DELETE**

REMOVE menu item

END IF

STORE update menu information

GENERATE update confirmation

SEND confirmation to Admin

END Update Meny Module

INPUT PARAMETERS:

- Menu Item Details (Name, Price, Category, Availability)
- Menu Update Type (Add / Update / Delete)
- Admin Identifier

OUTPUT PARAMETERS:

- Menu Update Confirmation Message
- Update Menu Information

GLOBAL VARIABLES:

- Menu Categories
- Restaurant Menu Policy

LOCAL VARIABLES:

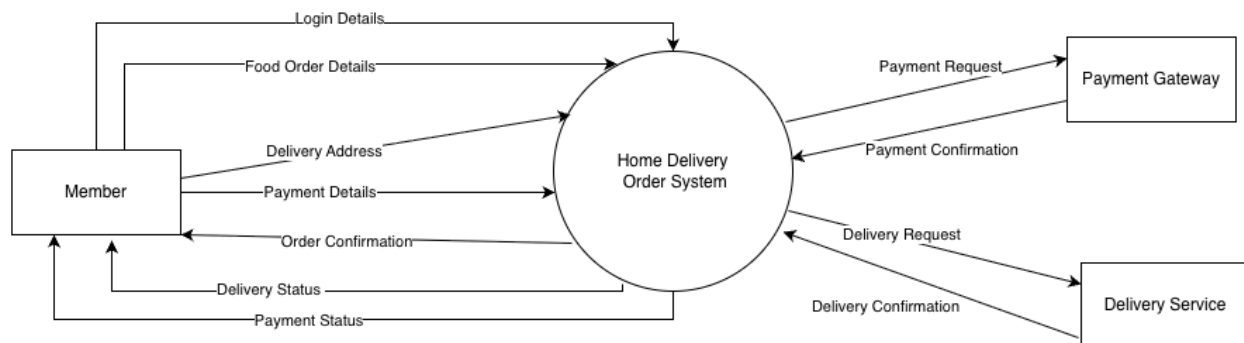
- Meny Item ID
- Update Status
- Validation Result

CALLS:

- Validate Menu Details Module
- Add Menu Item Module
- Update Menu Item Module
- Delete Menu Item Module
- Date Access Component: Menu Data Access Service

CALLED BY:

- Admin Interface
- Restaurant Management System (Main Controller)

10.9 Member 5:**Student Name: Inesh Puri****LMU ID: 24058811****Function Chosen: Place Order for Home Delivery****Context Level Diagram:***Figure 22 Home Delivery DFD Level 0***DFD LEVEL-1:**

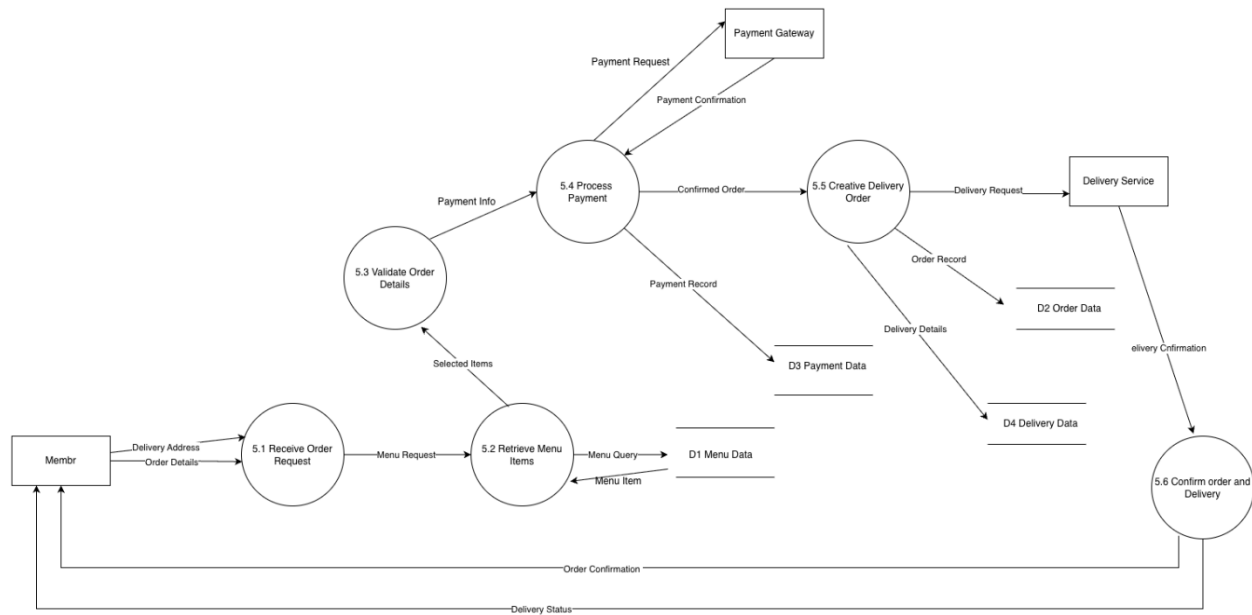


Figure 23 Home Delivery DFD Level 1

DFD LEVEL-2 (For Specification):

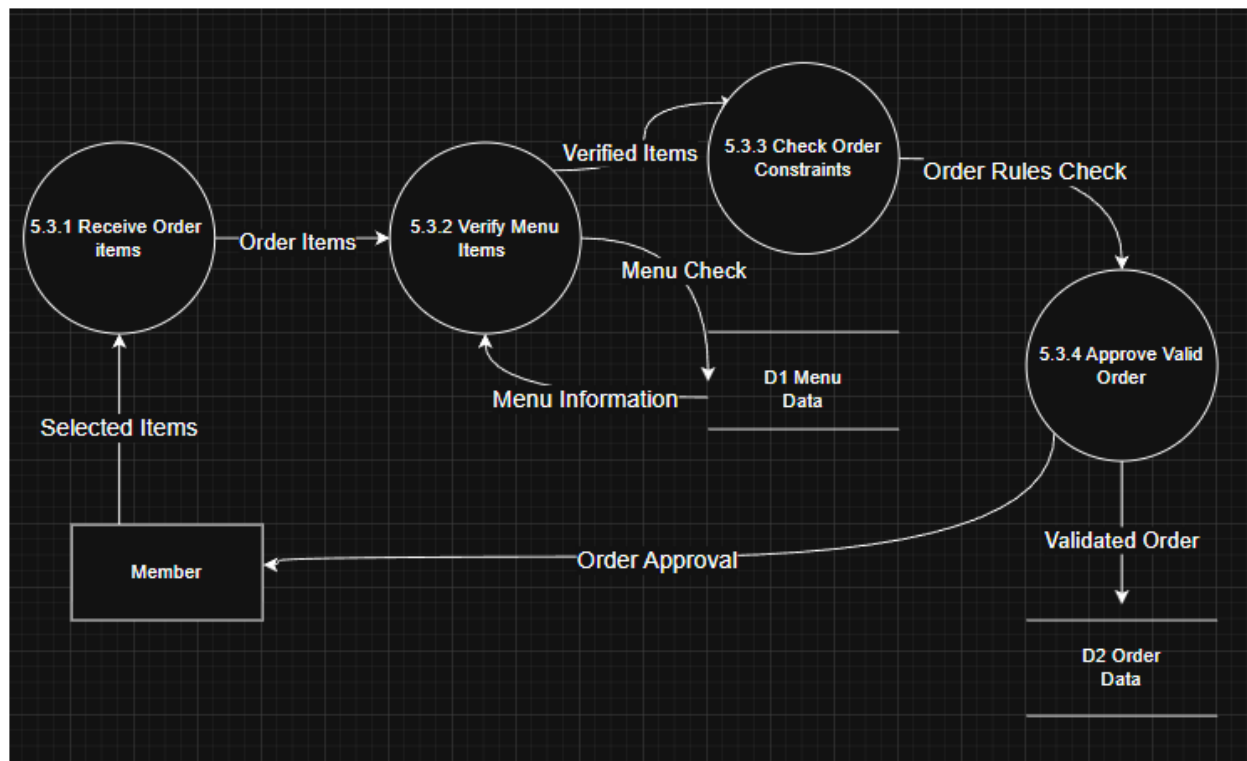


Figure 24 Home Delivery DFD Level 2

Design Specification:

Structure Chart for the Particular Function:**Module Specifications (MSpecs):**

MODULE NAME: Place Order for Home Delivery Module

PURPOSE: The Purpose of the Place Order for Home Delivery module is to allow registered members to place food orders for delivery to their specified address. This module handles receiving orders details, validating selected menu items, processing payment, creating delivery orders, and confirming the order and delivery status to the member. The module ensures accurate order processing and smooth coordination with payment and delivery services.

PSEUDOCODE:

START Place Order for Home Delivery Module

RECEIVE order request from Member

CAPTURE selected items and delivery address

RETRIVE menu details

VALIDATE selected items against menu

IF selected items are invalid

RETURN validation error message

END IF

GENERATE payment request

SEND payment request to Payment Gateway

RECEIVE payment confirmation

IF payment is successful

STORE payment details

CREATE order record

SEDN delivery request to Delivery Service

RECEIVE delivery confirmation

SEND order and delivery confirmation to Member

ELSE

RETURN payment failure message

END IF

END Place Order for Home Delivery Module

INPUT PARAMETERS:

- Selected Meny Items
- Delivery Address
- Payment Details
- Member Identifier

OUTPUT PARAMETERS:

- Order Confirmation Message
- Delivery Status Information
- Payment Status

GLOBAL VARIABLES:

- Delivery Charges
- Minimum Order Amount
- Restaurant Operating Area

LOCAL VARIABLES:

- Order ID
- Payment Transaction ID
- Delivery Request ID
- Order Status

CALLS:

- Retrieve Menu Items Module
- Validate Order Details Module
- Process Payment Module
- Create Delivery Order Module
- External Service: Payment Gateway API and Delivery Service API

CALLED BY:

- Member Interface
- Restaurant Management System (Main Controller)

11 Conclusion:

The proposed system addresses key operational challenges faced by modern restaurants, including table reservations with advance payment, cancellation and refund handling, home delivery ordering, menu and offer management, billing, reporting, and customer feedback. The separation of user's roles into Admin, Visitors, and Members ensures controlled access and improves system security and usability. Functional requirements clearly define what the system must do, while non-functional requirements specify performance, security, reliability, and maintainability constraints, ensuring the system is both efficient and robust. By applying logical modeling techniques such as DFDs and ERDs, the design minimizes data redundancy and consistency across transactions, payments and reports. The use of process specifications further clarifies internal system behavior, making the system design easier to understand, implement, and maintain. Additionally, the SRS provides a solid foundation for future development, testing, and evaluation of the system.

12 References

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